

02: Sampling

Stat 120 | Fall 2025

[List your group here]

To get started:

1. Click the “code” button
2. Click “copy to clipboard” button
3. In maize, open a new rmarkdown file (New file -> Rmarkdown -> (name the file 02-sampling.Rmd))
4. Paste the code into the file that opened up and hit “save”

Once you have the file loaded into RStudio, you can follow the directions below.

Press the “play” button below to run the chunk of code. This (1) loads the libraries that we need and (2) tells R to read `mission_data.csv` from my website folder into your R session and call it `mission_data`.

```
library(tidyverse)
mission_data = read_csv("https://www.math.carleton.edu/aluby/data/mission_data.csv")
```

Check your “environment” pane in the upper right to make sure you can see a dataset called `mission_data`. Try clicking it, or running `View(mission_data)` in the console to bring up the data viewer.

spoiler alert: The next chunk of code computes the *population mean*.

```
mean(mission_data$length)
```

```
[1] 5.947802
```

The code below selects a random sample of word positions to use

```
set.seed(091824) # Sets the random seed so we all get the same answer
sample = sample(1:nrow(mission_data), size = 10) # selects a random sample of size 10 from the mission_data
sample # prints the random sample of numbers
```

```
[1] 213 328 237 164 34 311 115 121 253 250
```

The next chunk of code `slices` our population to draw our sample. Note that the `position` variable should match the `sample` output above.

```
mission_sample = mission_data %>%  
  slice(sample)  
  
mission_sample
```

```
# A tibble: 10 x 4  
  paragraph word      position length  
    <dbl> <chr>    <dbl>   <dbl>  
1         3 be        213     2  
2         4 of        328     2  
3         4 the       237     3  
4         3 carleton  164     8  
5         1 learning   34     8  
6         4 for       311     3  
7         3 the       115     3  
8         3 students  121     8  
9         4 arts      253     4  
10        4 in       250     2
```

And this next chunk computes the mean word length of our sample

```
mean(mission_sample$length)
```

```
[1] 4.3
```

To try other random samples, change (or remove!) the `set.seed()` line of code, and try re-running the rest of the code. Do you ever get a sample mean that looks like your “by hand” sample mean?

When you’re done, **knit this file** and try uploading the PDF to gradescope. There are two questions, one for the *population mean* and one for a *sample mean*. Be sure to mark the pages so I can see your answers!